

Customer Shopping Behavior Analysis Dashboard

Project Overview

This project focuses on analyzing customer shopping behavior data using Python, SQL Server, and Power BI. The objective was to transform raw customer transaction data into meaningful business insights through data cleaning, exploratory data analysis (EDA), SQL-based business queries, and an interactive Power BI dashboard.

The project demonstrates end-to-end Data Analytics workflow including data preprocessing, feature engineering, database analysis, and business intelligence reporting.

Dataset Summary

The dataset contains customer shopping behavior records collected from an e-commerce retail environment. It captures customer demographics, purchasing patterns, product preferences, subscription information, shipping choices, and transaction details.

Dataset Statistics

Metric	Value
Total Records	3,900
Total Features	18
Missing Values	37
Duplicate Records	0
Product Categories	4
Unique Products	25
Locations Covered	50
Average Customer Age	44.1 Years
Average Purchase Amount	\$59.76
Total Revenue Represented	\$233,081

Features Included

Customer Information

- Customer ID
- Age
- Gender
- Location

Product Information

- Item Purchased
- Category
- Size
- Color
- Season

Transaction Information

- Purchase Amount (USD)
- Review Rating
- Previous Purchases
- Frequency of Purchases

Customer Engagement

- Subscription Status
- Discount Applied
- Promo Code Used

Logistics & Payment

- Shipping Type
- Payment Method

Dataset Objective

The dataset is designed to help analyze:

- Customer purchasing behavior

- Product performance
- Revenue trends
- Customer loyalty patterns
- Subscription impact on sales
- Discount effectiveness
- Shipping preferences
- Customer segmentation strategies

This dataset serves as an excellent case study for Data Cleaning, Exploratory Data Analysis (EDA), SQL Analytics, Business Intelligence Reporting, and Dashboard Development.

Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- SQL Server Management Studio (SSMS)
- T-SQL
- Power BI
- Google Colab

Dataset Description

The dataset contains customer shopping information including:

- Customer ID
- Age
- Gender
- Category
- Item Purchased
- Purchase Amount
- Review Rating
- Subscription Status

- Shipping Type
 - Discount Applied
 - Previous Purchases
 - Frequency of Purchases
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Data Cleaning & Preprocessing

The following preprocessing steps were performed:

Handling Missing Values

- Missing values in Review Rating were filled using the median value within each product category.

Feature Engineering

Created new features:

Age Group

Customers were segmented into:

- Young Adult
- Adult
- Middle-aged
- Senior

Purchase Frequency Days

Converted purchase frequency categories into numerical values:

Frequency	Days
Weekly	7
Fortnightly	14
Monthly	30
Quarterly	90
Annually	365

Data Standardization

- Converted column names to lowercase.
- Replaced spaces with underscores.
- Standardized naming conventions.

Redundant Column Removal

- Removed Promo Code Used column after validating that it duplicated Discount Applied information.
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SQL Business Analysis

The following business questions were solved using SQL:

Revenue Analysis

- Total revenue generated by Male vs Female customers.

Customer Spending Analysis

- Customers who used discounts and still spent above average.

Product Performance

- Top 5 products with highest average review ratings.

Shipping Analysis

- Comparison of average purchase amount between Standard and Express shipping.

Subscription Analysis

- Average spend and total revenue of subscribed vs non-subscribed customers.

Discount Effectiveness

- Top 5 products with highest percentage of discounted purchases.

Customer Segmentation

Customers categorized into:

- New Customers
- Returning Customers
- Loyal Customers

based on previous purchase count.

Category Analysis

- Top 3 most purchased products within each category.

Repeat Buyer Analysis

- Relationship between repeat purchases and subscription status.

Age Group Revenue Analysis

- Revenue contribution of different age groups.
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Power BI Dashboard

The dashboard provides interactive visualizations including:

Sales Overview

- Total Revenue
- Average Purchase Amount
- Total Customers

Customer Insights

- Revenue by Gender
- Revenue by Age Group
- Customer Segmentation

Product Insights

- Top Purchased Products
- Highest Rated Products
- Category Performance

Subscription Insights

- Subscriber vs Non-Subscriber Revenue
- Customer Spending Comparison

Discount Analysis

- Discount Usage Trends
- Discount Impact on Revenue

Shipping Analysis

- Purchase Amount by Shipping Type
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Key Business Insights

- Customer subscription status significantly impacts total revenue generation.
- Certain products achieve high customer ratings despite frequent discount usage.
- Loyal customers contribute a major share of purchases.
- Revenue distribution varies across age groups.
- Shipping preference influences customer spending behavior.

Project Workflow

Raw Dataset ↓ Data Cleaning (Python) ↓ Feature Engineering ↓ Exploratory Data Analysis ↓ SQL Business Queries ↓ Power BI Dashboard ↓ Business Insights & Recommendations

Project Outcome

This project demonstrates practical skills in:

- Data Cleaning
- Data Transformation
- SQL Query Writing
- Business Problem Solving
- Data Visualization
- Dashboard Development
- Business Insight Generation

The project successfully converts raw shopping behavior data into actionable insights that can support marketing, customer retention, product strategy, and revenue optimization decisions.

Author

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GitHub Portfolio Project – Customer Shopping Behavior Analysis Dashboard